PRANAY SHARMA

pshar147@asu.edu | +1(602) 384-7327 | linkedin.com/in/pranaysharma13 | https://pranaysharma13.github.io

EDUCATION

ARIZONA STATE UNIVERSITY Aug 2023- Present

Bachelor of Science in Computer Science | Minor in Business

Relevant Coursework: Intro to AI, Computer Networks, Intro to Financial Engineering, Linear Algebra, Information Assurance, Data Structures and Algorithms, Logic in Computer Science, Discrete Mathematical Structures, Probability and Statistics, Applied Data Analytics, Data Analytics with Tableau, Theoretical Computer Science, Operating Systems, Principles of Programming Languages, Macroeconomics, Microeconomics, Accounting, Marketing, Management & Strategy.

SKILLS

Languages: Python, SQL (MySQL), Java, C/C++, Prolog, MATLAB, Scheme, Dart

Developer Tools: Git, MS Office, VS Code, PyCharm, IntelliJ, Tableau

Frameworks: Django, Flutter

Soft Skills: Leadership, Management, Technical Writing, Social Media Management, Research, Communication

PROJECTS

PREMIER LEAGUE MATCH PREDICTION MODEL | *Python, NumPy, SciPy, JavaScript, Three.js*

- Built a Dixon-Coles bivariate Poisson model from scratch with a hand-derived analytic gradient, fitting team strength ratings via weighted maximum likelihood across 6,600+ matches and two divisions (sub-second fit time).
- Extended the likelihood to a mixed Poisson/Gamma formulation to incorporate expected-goals (xG) data alongside match results, improving out-of-sample prediction accuracy (RPS) by 1-4% across three seasons of walk-forward backtesting.
- Validated the model against closing bookmaker odds as a benchmark, quantifying exactly where the model added value and where it did not (e.g., zero blending weight vs. market odds- a rigorous negative result).
- Diagnosed and corrected systematic overconfidence in season-long Monte Carlo projections (68.6% actual vs. 80% target coverage) through empirical calibration against 7 seasons of historical outcomes.
- Shipped a zero-backend, static-data dashboard (vanilla JS + Three.js) visualizing 20,000-simulation season projections, including an interactive match predictor computing live probabilities client-side.

DNA SEQUENCE NFA PATTERN MATCHER | *Python, Streamlit, NetworkX*

- Implemented Thompson's Construction from scratch via a recursive descent parser, converting DNA regex patterns into NFAs without any automata libraries.
- Built ε-closure NFA simulation engine supporting wildcards, alternation, grouping, and character classes over the {A, C, G, T} alphabet.
- Developed subset construction (NFA to DFA) algorithm, visualizing state-set correspondence and worst-case 2^n state blowup interactively.
- Delivered a 5-tab Streamlit dashboard with live NFA/DFA graph diagrams, step-by-step execution animation, and highlighted match results for biological patterns including start codons, stop codons, and CpG islands.

DATA ANALYSIS FOR SUSTAINABILITY | *Tableau*

- Worked on real-world energy data in collaboration with Intel Corporation’s Sustainability Team.
- Built interactive dashboards in Tableau using calculated fields, parameters, filters, and dual-axis charts.
- Analyzed KPIs such as Net Energy Production, Renewable Energy Share, and Hourly Generation Changes.
- Recommended optimal U.S. region for Intel’s new data center based on multi-factor energy analysis.

GRAMMYS WEBSITE AUDIENCE ANALYSIS | *Microsoft Excel*

- Prepared real data provided by The Recording Academy by transforming it.
- Calculated Key Business Performance Indicators (KPIs) and provided recommendations to the organization.

- Project aligns with United Nations SDG 9: Industry, Innovation, and Infrastructure.

AUDIBLE ASU MAPS FOR WAYFINDING | *Dart, Flutter*

- Developed a cross-platform audio input/output- enabled navigation app using Dart and Flutter, to assist visually impaired students at ASU, under the guidance of Dr. Jared Schoepf.
- The app features live location tracking, building detection of up to 35 m radius, walking ETA, and waypoint polyline graphs.
- The team presented this project to a jury and were appreciated for the work.

EXPERIENCE

TECHNOLOGY CONSULTANT, ENTERPRISE TECHNOLOGY, ASU *Tempe, AZ* Sept 2026- Present

- Deliver in-person technical support for personal devices, university software, and general technology issues for students, faculty, and staff.
- Diagnose and troubleshoot hardware, software, account, and connectivity problems while providing excellent customer service.
- Assist with device setup, software installation, and technology guidance, escalating complex issues as needed.
- Collaborate with team members to provide efficient walk-in technical support in a fast-paced service environment.

SECTION LEADER, IRA A. FULTON SCHOOLS OF ENGINEERING, *Tempe, AZ* Aug 2024- Dec 2024

- ASU 101 Section Leader for a class of 19 students under Professor Steven Osburn.
- Delivered engaging lectures, organized a library tour, evaluated assignments, and facilitated one-to-one student mentoring.
- Provided academic guidance and connected students with institutional resources to support their transition to university life.
- Coordinated communication between the professor and department, including class updates and attendance reporting.

SOFTWARE ENGINEERING FELLOW, HEADSTARTER AI, *United States (Remote)* July 2024- Aug 2024

- Worked towards the Track A objective of accumulating 1000 waitlist users, creating 1000 user accounts, or generating revenue exceeding $1000 through innovative AI projects.
- Engaged in extensive networking with industry professionals and peers to enhance collaboration and learning.

STUDENT MEMBER, FACE, *United States (Remote)* Aug 2023- May 2024

- Identified, explored and applied my top five Clifton Strengths.
- Conducted interviews with a Fulton leader to gather experiential insights and researched on Diversity, Equity, and Inclusion.
- Participated in Fulton Student Organization meetings and documented them using a qualitative observation approach.
- Developed a SMART goal and collaborated on a project addressing groundwater depletion in California’s Central Valley.
- Presented my yearlong work at the FACE symposium to the engineering faculty.